Mradul Sonkar

B.Tech. CSE, IIT Bombay
sonkarmradul@gmail.com
+91-6375549617 |

Scholastic Achievements

- Achieved an **All India Rank of 1039** out of over **180,000** candidates in the JEE Advanced examination *(2023)*
- Achieved **99.96 percentile** with an **All India Rank of 596** in JEE Main out of **1.1 million** candidates *(2023)*
- Ranked among the **National top 1%** in **NSEC** (National Standard Examination in Chemistry) by IAPT *(2022)*
- Secured an **All India Rank of 833** among **90,000+** eligible students in **KVPY** examination held by IISc *(2021)*
- Awarded the National Talent Search Examination (**NTSE**) scholarship by **NCERT**, Government of India *(2021)*

Professional Experience

Product Intern | *Adobe* *(May'26 – July'26)*
Summer Internship

- Built an **LLM-driven agent** to automate crash/ANR root-cause analysis for Adobe Express, cutting the triage time from **2-3 hours** of manual New Relic and Splunk digging down to **under 10 minutes** (**∼90%+** reduction)
- Designed a **multi-skill pipeline**, integrating **5 MCP servers** for multiple platforms like New Relic and Splunk
- Optimized the agent's pipeline, separating context-fetching from reasoning, reducing per analysis cost by **∼98%**
- Framed an **autonomous crash-reproduction pipeline** that drives appropriate Android emulator image and uses a **screen-perceive-act feedback loop** to verify hypothesized root causes and provide exact reproduction steps

Frontend Intern | *Tria* *(December'25 - April'26)*
Winter Internship

- Built **3** live internal dashboards using **Next.js** for the frontend and **Node/Express** REST APIs for the backend
- Designed backend APIs to query and aggregate data across multiple **SQL** tables to calculate the dashboard metrics
- Structured **SQL** queries around metric groups instead of one join, keeping slow metrics from bottlenecking the rest

Key Projects

Mobile App for NSS IITB | *Technical Project* *(December'24 – March'25)*
Web Department, National Service Scheme | IIT Bombay

- Built an **app** for **450+** NSS IITB volunteers using **Flutter** for frontend and backend with **Express.js** and **SQL**
- Implemented a secure **role-based authentication**, enabling differentiated access control for volunteers and admins
- Developed essential features like **registration**, **progress tracking**, **attendance marking** and **events management**
- Drove the development in a **4**-member team, using **Git** and **GitHub** for version control and organized workflows

C Compiler Development | *Course Project* *(January'26 – April'26)*
Instructor: Prof. Uday Khedkar | Course: CS320

- Built a C compiler using **Lex** for **lexical analysis** and **Yacc** for **parsing**, translating C into intermediate codes
- Designed **symbol table management** and construction of **Abstract Syntax Trees** for a simplified C language
- Generated **Three Address Code (TAC)**, **Register Transfer Language (RTL)** and **MIPS assembly code**

xv6 Kernel Enhancement | *Course Project* *(January'25 – April'25)*
Instructor: Prof. Mythili Vutukuru | Course: CS236

- Extended the **xv6 kernel** by implementing more than **10 system calls** and building a custom **user-level shell**
- Replaced the default process scheduler with a **priority-based round-robin scheduler**, improving process handling
- Explored IPC mechanisms such as **pipes**, **signals**, and **shared memory** to enhance inter-process communication
- Gained in-depth exposure to **multiprocessing, synchronization**, and **user-level thread management** in OS

Plagiarism Detector | *Course Project* *(August'24 – November'24)*
Instructor: Ashutosh Gupta | Course: CS213

- Developed a plagiarism detector using **KMP** algorithm for efficient string matching, ensuring precise content analysis
- Implemented advanced **multithreading** to enable simultaneous file checks, optimizing speed and resource utilization
- Applied **mutex locks** for **thread safety**, ensuring consistent and secure execution in a **concurrent** environment

MyGit: Version Control System | *Personal Project* *(May'24 – July'24)*
A lightweight version control system | Inspired by Git

- Built a custom **version control system** supporting commands like `init`, `add`, `commit`, `checkout`, `log` and `branch`
- Employed **file manipulation** to manage snapshots of project and used **SHA-1 hashing** to detect changes efficiently
- Created a structured internal metadata system to log commits, maintain pointers, and facilitate version traversal

Hand Gesture Recognition | *Course Project* *(August'25 – November'25)*

Instructor: Prof. Suyash Awate | Course: CS663

- Compared **SVM**, **CNN** and **MediaPipe** landmark-based models for **hand gesture recognition** across **36** classes
- Built a pipeline with **Otsu thresholding**, **LoG** contours and **Hu moments**, training an **SVM** to **80.7%** accuracy
- Built a **CNN** based classifier reaching **95%** test accuracy and analyzed its overfitting to background and skin-tone
- Extracted 21 **MediaPipe** hand landmarks, normalized them and trained a **dense neural network** for predictions

Stock Market Predictor | *Course Project* *(January'24 – April'24)*

Instructor: Pushpak Bhattacharyya | Course: CS240

- Built a **stock predictor** using **PyTorch** and **deep-river**, with **past prices** and **volume** as inputs to the model
- Wrapped **LSTM model** inside **RollingRegressor** to enable **online learning** with continuously updating weights
- Achieved a **MAE** score of **11.96** (∼**1%** error) on live close-price predictions, validated across **10** NSE-listed stocks
- Built an intuitive **UI** using **Streamlit**, offering interactive access to model predictions and **live visual feedback**

VoxElectra: Audio Transcription with Transformers | *Seasons Of Code* *(May'24 – July'24)*

Web and Coding Club | IIT Bombay

- Built an end-to-end **CNN** + **BiGRU** speech-to-text pipeline on **LJSpeech** spectrograms, trained with **CTC loss**
- Trained a **CNN** on **MFCC** audio features to classify 8 spoken voice commands achieving **93.5%** validation accuracy
- Doubled training data via **noise-injection augmentation**, benchmarking transcription accuracy with **WER**

Positions Of Responsibility

Web Head | *National Service Scheme, IIT Bombay* *(April'25 – March'26)*

- Led the **Web Department** at **NSS IITB**, managing a team of **3** Activity Associates and all key tech initiatives
- Oversaw the maintenance and development of **institute-wide portals** to streamline core organizational workflows
- Revamped the official website of NSS IITB using **React** and **Tailwind CSS** for better **scalability** and **performance**
- Oversaw and developed final features of the **NSS IITB app**, including attendance marking and event management

Web Activity Associate | *National Service Scheme, IIT Bombay* *(May'24 – March'25)*

- **Ideated**, **developed** and **deployed institute-wide** websites for NSS IITB as a part of 4-member web team
- Developed and deployed **responsive websites** with **clean** and **intuitive UIs** for every sub-department of NSS
- Spearheaded the development of the **NSS IITB app** for **attendance tracking**, **event updates**, **notifications**

Technical Skills

Programming Languages	C++, Python, JavaScript, TypeScript, Bash, MIPS32 Assembly, Markdown, SQL
Development and Tools	HTML, Tailwind CSS, ReactJS, Express.js, NodeJS, Flutter, LaTeX, Git, GitHub
Python Libraries	NumPy, Matplotlib, Pandas, Seaborn, Librosa, scikit-learn, TensorFlow, PyTorch

Courses Undertaken

Computer Science	Data Structures and Algorithms†, Artificial Intelligence and Machine Learning†, Design and Analysis of Algorithms, Operating Systems†, Computer Architecture†, Data Analysis and Interpretation, Software Systems Lab, Discrete Structures, Logic for Computer Science, Abstractions and Paradigms for Programming†, Computer Programming and Utilization, Web Security†, Computer Networks†, Implementation of Programming Languages†, Database and Information Systems†,
Mathematics & Others	Calculus, Linear Algebra and Differential Equations, Economics, Introduction to Entrepreneurship, Physics Lab, Chemistry Lab, Makerspace, Design Thinking

†*Course has corresponding lab*

Extracurricular Activities

- Participated and completed the **WnCC x GFG DSA Bootcamp**, competing against **300**+ students *(2025)*
- Completed **Deep Learning Bootcamp** (**Natural Language Processing** track) by Analytics Club, IITB *(2024)*
- Mentored 7 mentees in **DSA** reading project under **Summer of Science'25**, covering core data structures *(2025)*
- Supervised the development of a custom **version control system**, inspired by **Git**, during **SoC'25** program *(2025)*
- Mentored students in building a **speech-to-text transcriptor** using **transformers** during **WiDS'24** *(2024)*